DS701 Homework 9: Causal Inference and Neural Networks

Released Mon Nov 9 · due Mon Nov 16, 11:59 pm on Gradescope

Name: _____ **BU ID:** _____

How this works. Answer **by hand on paper** (or a tablet with a stylus), show your work, then scan and upload to Gradescope, matching each problem to its pages. This is deliberate practice for the closed-notes quizzes, which are in the same style — so do it as much as possible without a computer or an AI assistant. If you are stuck, you can try AI in tutorial style or reach out to course staff. It is **participation-graded**: a serious attempt at every problem earns full credit, and worked solutions are released the day after the deadline. Seven short parts; budget 1–2 hours.

Covers: L14 Causal Inference (Mon Nov 9), L16 Neural Networks (Wed Nov 11). **Reference sheet:** quiz-3 sheet §14 and §16 — allowed and encouraged; every part is answerable from it plus what is printed.

Problem 1: Does the coupon work? (read the table)

Practices: crude vs. stratified comparisons, Simpson's paradox, and one standardization (L14).

A streaming service e-mailed a "welcome back" coupon to some lapsed subscribers: $T = 1$ if a user got the coupon, $Y = 1$ if they resubscribed within 30 days, and $Z = 1$ if the account had been lapsed **more than a year** ("long-lapsed"). Counts for 400 users; work with resubscription *rates* and the **risk difference** $P(Y{=}1 \mid T{=}1) - P(Y{=}1 \mid T{=}0)$ as the measure of effect:

	long-lapsed (Z=1)		recently lapsed (Z=0)	
	users	resubscribed	users	resubscribed
coupon (T=1)	260	52	40	32
no coupon (T=0)	40	4	60	42

(a) Compute the **crude** (pooled) resubscription rate for coupon and no-coupon users and the crude risk difference, then the risk difference **within** each stratum of Z. Name the phenomenon, name the role Z plays, and point to the **two ratios in the table** that are its fingerprints.

Crude rate, coupon: _____ no coupon: _____ crude difference: _____
Difference within Z=1: _____ within Z=0: _____

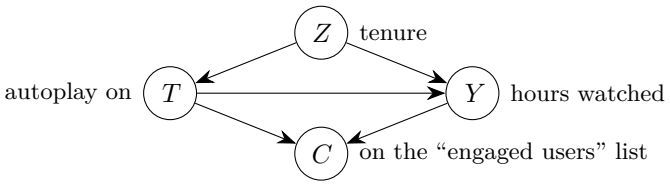
(b) Use the sheet's adjustment (standardization) formula $P(Y \mid do(T{=}t)) = \sum_z P(Y \mid T{=}t, Z{=}z)\, P(Z{=}z)$ to compute $P(Y{=}1 \mid do(T{=}1))$, $P(Y{=}1 \mid do(T{=}0))$ and their difference. In one sentence: what does this number estimate that the crude difference does not?

$P(Y{=}1 \mid do(T{=}1))$ = _____; $P(Y{=}1 \mid do(T{=}0))$ = _____; difference = _____

Problem 2: One small DAG

Practices: reading a causal DAG — fork vs. collider — choosing the one variable to adjust for, and prediction vs. intervention (L14).

The same service wants the **total effect** of its **autoplay** setting ($T = 1$ if on) on **weekly hours watched** Y. Its analysts agree on this DAG: long-tenured subscribers (Z = tenure) are more likely to leave autoplay on *and* watch more regardless, and marketing keeps a list C of "engaged users" — anyone with autoplay on **or** many hours watched. Because that list has the cleanest data, one analyst proposes comparing autoplay-on and autoplay-off users **within the list only**.



(a) To estimate the total effect of T on Y there is exactly **one** variable in this DAG you should adjust for. Which one, and why — name the building block (chain / fork / collider) that Z and C each form with T and Y, and the sheet's rule for each. Then, in one or two sentences: what goes wrong with the "within the list only" proposal?

Adjust for: _____ Z is a: _____ C is a: _____

(b) **True or false, with one reason each** (a sentence is enough):

(i) A model that predicts hours watched very accurately from the autoplay setting and other variables therefore tells us how much hours watched would change if the service switched autoplay on for a user.

(ii) If the service randomly assigned autoplay on/off to new accounts, the plain difference in mean hours between the two arms would be an unbiased estimate of the average treatment effect, *even though tenure Z is not measured.*

(i) True / False, because:

(ii) True / False, because:

Problem 3: Training a small network

Practices: what the parameters of an MLP are, reading training and validation loss, and one gradient-descent step (L16).

A team fits a regression MLP with **6 (scaled) input features**, **one hidden layer of 8 ReLU units**, and **one linear output** $\hat{y}$, trained with mean squared error.

(a) How many trainable parameters does the network have? Show the count as "weights + biases" for each layer (every entry of the sheet's Ω_k and β_k counts). A teammate then suggests removing the ReLU ("the hidden layer is still there, so we keep the same capacity"). **True or false**, with one reason — and say how many *effective* parameters the ReLU-free network has.

Total parameters: _____ ReLU-free claim is: True / False
Effective parameters without ReLU: _____

(b) Three training runs on the same data. Run A and Run B use the architecture above and differ only in `learning_rate_init`. Run C uses Run A's settings but the team forgot the `StandardScaler`, so feature 3 (annual income, in dollars) enters raw. Losses per epoch (Run C's row is in units of 10^8):

epoch	1	2	3	4	5	6	7	8	9	10
Run A ($\eta = 0.01$), train	0.92	0.61	0.44	0.35	0.29	0.25	0.22	0.19	0.17	0.15
Run A, validation	0.95	0.66	0.51	0.44	0.40	**0.38**	0.39	0.41	0.44	0.48
Run B ($\eta = 1.0$), train	0.92	1.61	0.88	2.73	5.40	4.12	9.85	21.7	18.3	47.6
Run C (unscaled), train $/10^8$	2.1	2.0	2.0	1.9	1.9	1.9	1.9	1.9	1.9	1.9

(i) Diagnose **Run B** in one sentence, and name the knob and the direction you would turn it.

(ii) In **Run A**, what is happening from epoch 7 on, and what is it called? The team re-runs A with `early_stopping=True` and `n_iter_no_change=3`, which here means: *stop as soon as the validation loss has failed to improve on its best value for 3 consecutive epochs, and keep the weights from the best epoch.* After which epoch does training stop, and which epoch's weights are kept?

(iii) Diagnose **Run C** in one sentence: why does one raw feature make gradient descent behave like this, and what is the fix?

(i)

(ii) this is called: _____ stops after epoch: _____ weights kept from epoch: _____

(iii)

(c) At some step of Run A, one hidden unit has incoming weights $\mathbf{w} = (0.40, -0.90, 1.50)$ and bias $b = 0.20$, and backprop has already delivered the gradients $\partial L/\partial \mathbf{w} = (-0.50, 0.20, 2.00)$ and $\partial L/\partial b = -1.00$. Apply **one** gradient-descent update with $\eta = 0.1$ and write the new $\mathbf{w}$ and b. Then: with $\eta = 1.0$ instead, what would the third weight become, and what does the sheet say a learning rate that large risks?

New $\mathbf{w} = ($ _____ , _____ , _____ $)$ new $b =$ _____ w_3 with $\eta = 1.0$: _____